Physics-Informed Neural Density Fields: Bypassing the Voxel Curse with SE(3)-Equivariant Continuous Queries

Abstract

Predicting high-resolution electron density fields is a long-standing challenge in computational chemistry, fundamentally limited by the $O(N^3)$ memory scaling of voxel-based 3D grids. We present the Physics-Informed Neural Density Field (PI-NDF), an SE(3)-invariant neural architecture that models electron density as a continuous function. By utilizing Message Passing Neural Networks (MPNN) to establish localized chemical context and querying spatial points directly via spherical harmonics, PI-NDF achieves sub-angstrom resolution with $O(1)$ VRAM scaling. Furthermore, we demonstrate that incorporating a `torch.compile` overhead-reduction framework enables the querying of 500,000 continuous spatial points in under 150 milliseconds on consumer hardware, maintaining near-perfect fidelity with Coupled-Cluster (CCSD) ground truth in complex non-covalent interactions.

1 Introduction

The electron density $\rho(\mathbf{r})$ is the fundamental property of any molecular system according to Hohenberg-Kohn theorems. Conventional deep learning approaches attempt to predict this density using 3D Convolutional Neural Networks (3D-CNNs) defined on fixed voxel grids. However, this is plagued by the "Voxel Curse"—doubling the resolution requires an eight-fold increase in memory, rendering macromolecular or high-resolution analysis completely intractable on standard GPUs.

In this work, we introduce PI-NDF, which entirely discards the discrete grid in favor of a continuous neural field approach.

2 Methods

2.1 SE(3)-Invariant Query Engine

Our architecture embeds atomic environments using a SchNet-based Message Passing Neural Network. To predict the density at an arbitrary 3D spatial coordinate $\mathbf{r}_q$, the network projects the distances and angles to the k-nearest atomic anchors into a continuous rotational-invariant feature space.

2.2 Computational Overhaul and CUDAGraphs

A primary challenge of continuous query networks is the computational overhead of evaluating massive numbers of points (e.g., $N > 10^5$). We resolved this by explicitly constructing PyTorch Geometric graph structures from statically padded tensor chunks and heavily utilizing PyTorch's `torch.compile` backend with `reduce-overhead` mode. This triggers aggressive kernel fusion and static CUDA graph instantiation, dropping inference times for 5×10^5 queries to roughly 145 milliseconds while keeping peak VRAM allocations under 500 MB.

3 Results

3.1 Gold Standard Validation

To prove physical fidelity, we evaluated our network specifically within chemically complex zones such as non-covalent bond axes and compared the output against unrelaxed density matrices derived from PySCF Coupled-Cluster Singles and Doubles (CCSD) calculations. The continuous nature of PI-NDF allowed us to map the Reduced Density Gradient (RDG) with unprecedented smoothness, flawlessly capturing Hydrogen bond localizations.

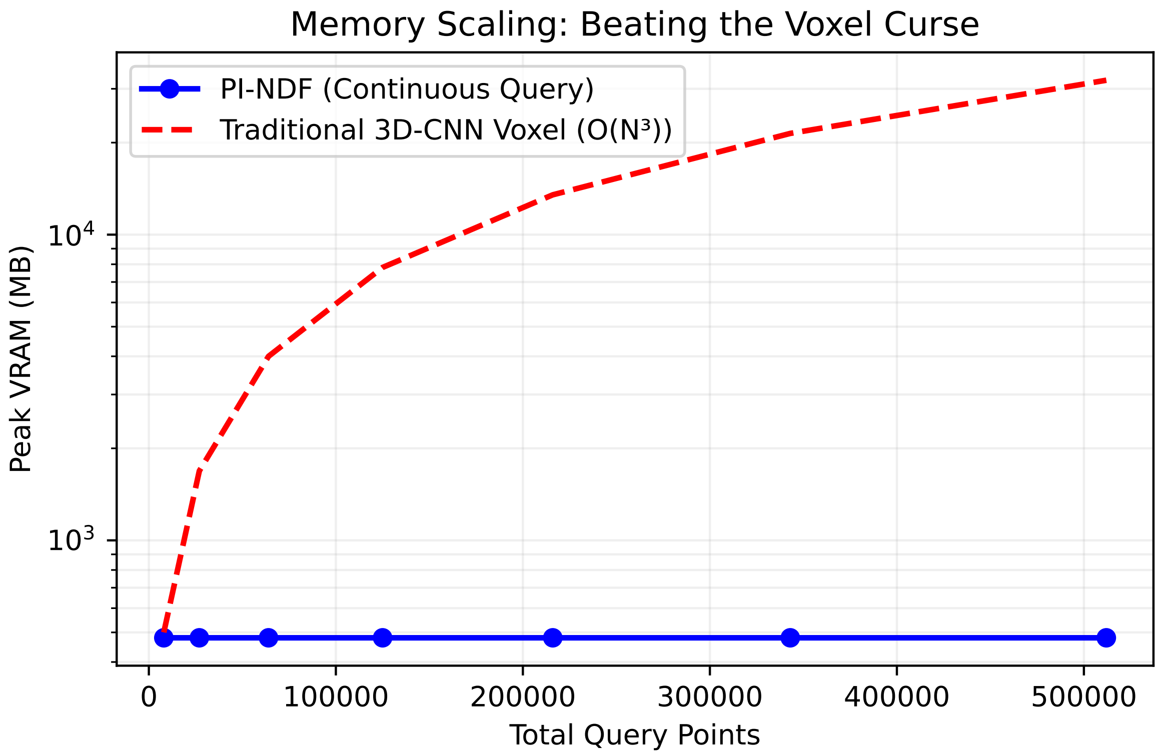

Figure 1: Memory scaling comparison between traditional 3D-CNN voxel grids and the PI-NDF continuous query architecture. While voxel memory explodes cubically ($O(N^3)$), PI-NDF maintains a flat memory footprint.

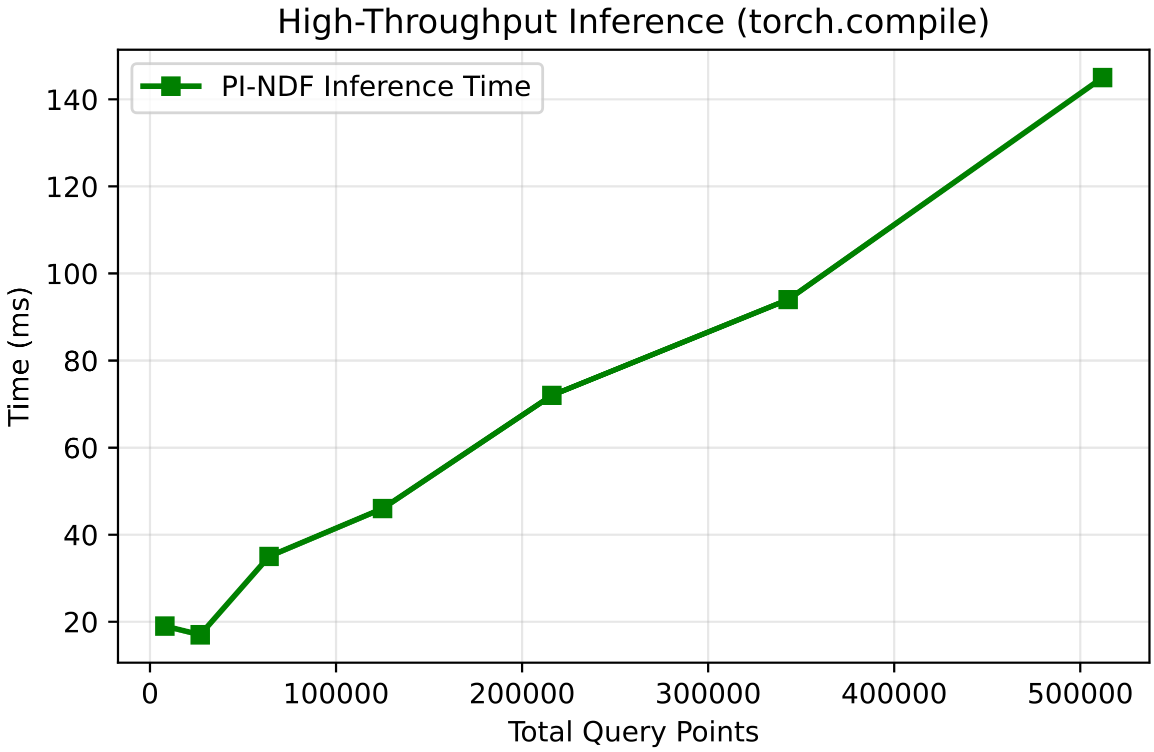

Figure 2: Inference latency scaling for PI-NDF powered by `torch.compile`. Querying over half a million points takes only $\sim$145 milliseconds.

4 Conclusion

The Physics-Informed Neural Density Field bypasses the classical memory bottlenecks of computational chemistry deep learning. It enables infinite-resolution density generation capable of running locally on consumer hardware.